MIL-L-40069D
4 November 1980
SUPERSEDING
MIL-L-40069C
20 January 1971

MILITARY SPECIFICATION

LEATHER, CATTLEHIDE, CHROME TANNED, COLD-CLIMATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers chrome tanned cattlehide leather (see 6.1).

1.2 Classification. The leather shall be furnished in the following types and classes as specified (see 6.2):

Type I - Sides
Type II - Backs
Type III - Bends
Type IV - Double shoulders

Class 1 - Mildew resistant treated
Class 2 - Not mildew resistant treated

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8330

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SPECIFICATION

MILITARY

MIL-L-43432 - Leather, Packaging and Packing Of.

STANDARD

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Unless otherwise specified (see 6.2), the leather shall be full grain or corrected grain cattlehide.

3.2 Tannage. The leather shall be chrome tanned.

3.3 Physical requirements.

3.3.1 Trim (form) and area. The trim of the sides, backs, bends or double shoulders shall conform with the definitions for sides, backs, bends, or double shoulders as listed under "Definition of Terms" in FED-STD-311. The area expressed in terms of square feet shall be legibly marked in the butt area on the flesh side of the leather except that double shoulders shall be marked near the head area.

3.3.2 Thickness. The thickness of the leather shall be as specified (see 6.2). A tolerance of $\pm$ 1/2 ounce shall be permitted (NOTE: 1 ounce = 1/64 inch).

3.3.3 Color. The color of the leather shall be as specified (see 6.2). The color may be applied by brush (surface) or drum dyeing. The color need not penetrate the entire thickness of the leather.

* 3.3.4 Finish. Unless otherwise specified (see 6.2), a finish shall not be applied to the leather. When a finish is specified, the color of the finish shall match the color of the dye. Unless otherwise specified (see 6.2), the leather shall be smooth plated. The color shall be uniform throughout the grain surface and the color shall be uniform throughout the flesh surface. The tanned leather shall be lubricated in such a manner as to provide the required flexibility under extreme cold conditions (see 6.4).

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3.3.5 Colorfastness. The leather shall be tested for resistance to wet and dry crocking in accordance with 4.3. The leather shall show fastness to crocking equal to or better than the standard sample (see 6.3). When no standard sample is available the staining of the dry cloth shall be not lower numerically than Munsell value 9.0 and staining of the wet cloth shall be not lower numerically than Munsell value 8.5. When the color of the leather is natural as produced by the tanning agents, the leather shall not be tested for crocking.

3.3.6 Breaking strength. At least 80 percent of the specimens tested shall meet the breaking force values specified in table I, when tested as specified in 4.2.2.2.

TABLE I. Breaking force

Thickness (ounces)	Breaking force, minimum (pounds)
2 to less than 5	30
5 to 6-1/2	70
7 to 8-1/2	125
9 and up	175

* 3.3.7 Elongation. At least 80 percent of the specimens tested shall have elongation values no greater than 25 percent when tested as specified in 4.2.2.2. For leather less than 5 ounces in thickness, the elongation shall be determined at a load of 20 pounds. For leather 5 ounces or greater in thickness, the elongation shall be determined at a load of 50 pounds.

3.3.8 Stiffness. The leather shall have a load-scale reading no greater than the values shown in table II when tested as specified in 4.2.2.2.

TABLE II. Stiffness

Thickness (ounces)	Load scale reading (maximum)
2 to 6-1/2	30.0
7 and up	60.0

3.3.9 Shrinkage. The leather shall not shrink before the temperature reaches 98°C nor shall the leather shrink when subjected to a temperature of 98°C ± 0.5°C for a period of 30 seconds when tested as specified in 4.2.2.2.

3.4 Chemical requirements. The leather shall conform to the requirements of table III, when tested as specified in 4.2.2.2.

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TABLE III. Chemical requirements

Characteristic	Minimum	Maximum
Chloroform soluble materials, percent	-	5.0
pH value	3.3	-
Chromic oxide, percent	3.0	-

3.5 Class 1. When class 1 is specified, the leather shall contain not less than 0.18 percent and not more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.2.2.2.

3.6 Workmanship. The finished leather shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Unless otherwise specified, inspection shall be performed in accordance with FED-STD-311.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.2.2 End item inspection.

4.2.2.1 End item examination. The defects found during the examination of the leather for visual and dimensional characteristics shall be in accordance with 4.2.2.1.1 and 4.2.2.1.2. The sample unit for these examinations shall be one side, back, bend, or double shoulder. The lot size shall be expressed in units of one side, back, bend, or double shoulder. Any sample unit containing one or

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more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with FED-STD-311 except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more units of product. Separate examinations shall be performed for visual and dimensional characteristics, but the same sample may be used for each examination.

4.2.2.1.1 Visual characteristics.

Examine	Defect
Trim	Not as specified.
Color	Not as specified.
Finish	Leather has application of finish (except when specified). Flesh or grain surface not uniform in color. Not smooth plated (or not embossed as required when specified).
Quality of leather	Not full grain or not corrected grain (see 6.2). Stain or other foreign matter. Any of the following defects which individually or collectively cause more than a 12 percent loss in the area of the leather: coarse or loose flesh, cut, scratch, hole, brand hip mark, rough shoulder, or other surface imperfections.

NOTE: Light well-healed scratches or grub holes, light fat wrinkles, or slight stains shall not be classified as defects.

4.2.2.1.2 Dimensional characteristics.

Examine	Defect
Area (expressed in terms of square feet)	Less than area marked on the leather.
Marking	Omitted.
Thickness 1/	Not as specified.

1/ Thickness shall be determined in accordance with Method 1011 of FED-STD-311, except that no individual measurement shall be less than the minimum or greater than the maximum specified.

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4.2.2.2 End item testing. The methods of testing specified in FED-STD-311, wherever applicable and as listed in table IV, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exists:

a. More than three test failures occur for breaking force or elongation.

b. More than one test failure occurs for the remaining requirements applicable to the sample unit.

c. Any composite test result fails to meet the standard requirement.

TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite sample
Material identification	3.1	1/	-	-
Tannage	3.2	1/	-	-
Colorfastness	3.3.5	3031	X	-
Breaking strength	3.3.6	2021	X	-
Elongation	3.3.7	2021	X	-
Stiffness	3.3.8	4.3.1	X	-
Shrinkage	3.3.9	7011	X	-
Chloroform-soluble material	3.4	6341	-	X
pH value	3.4	6621	-	X
Chromic oxide	3.4	6515	-	X
Paranitrophenol, class 1	3.5	6711	-	X

1/ A certificate of compliance shall be submitted and will be acceptable for the requirement stated.

4.2.3 Packaging inspection. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine that preservation, packing and marking comply with the section 5 requirements.

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4.3 Methods of inspection.

4.3.1 Stiffness. The stiffness of the specimen shall be determined as specified in Method 4211 of FED-STD-311, using a one inch span and a total moment weight equivalent to 0.5 inch-pound. The specimen shall be clamped firmly in the vise and the apparatus engaged until the specimen has been subjected to an angular deflection of 40 degrees. The apparatus shall be immediately readjusted by hand so as to completely remove the load on the specimen. Then the specimen shall be subjected to an angular deflection of 20 degrees following the procedure specified in Method 4211. The stiffness shall be recorded as the load scale reading.

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. Leather shall be preserved in accordance with the applicable requirements of MIL-L-43432.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. Leather shall be preserved and packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use. Class 2 leather is intended for use in clothing and equipage used under cold climate conditions. Class 1 leather is intended for use where such items might also be used under tropical conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and class required (see 1.2).
(c) Type of grain (either full grain or corrected grain) when a preference is desired (see 3.1).
(d) Thickness required (see 3.3.2).
(e) Color required (see 3.3.3).
(f) Finish, when required, and type of embossing when smooth plating is not required (see 3.3.4).
(g) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Samples. For access to sample, address the procuring activity issuing the invitation for bids.

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* 6.4 Flexibility of cold climate leather. Cold climate leather of the desired flexibility has been produced successfully by the following methods:

(a) By incorporating into the leather small amounts of fat-liquors or other lubricants, drying of the leather, and extraction of the dried leather with a fat solvent to reduce its chloroform-soluble materials to 5.0 percent or less. NOTE: Paramtrophenol is soluble in chloroform. Consideration should be given to this fact before extracting with chloroform.

(b) By incorporating into the leather, lubricants that do not appreciably stiffen at low temperatures.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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MIL-L-40069D Leather, Cattlehide, Chrome Tanned, Cold-Climate

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